

$$\mathbf{U} \cdot \mathbf{\Sigma} \cdot \mathbf{V} = \mathbf{Y}$$

The diagram illustrates the full Singular Value Decomposition (SVD) of matrix $\mathbf{Y}$. Matrix $\mathbf{U}$ is a 10x10 orthogonal matrix. Matrix $\mathbf{\Sigma}$ is a 10x10 diagonal matrix containing the singular values of $\mathbf{Y}$. Matrix $\mathbf{V}$ is a 10x10 orthogonal matrix. The product of these three matrices equals matrix $\mathbf{Y}$.

 Truncation based on $\mathbf{\Sigma}$

$$\mathbf{U}' \cdot \mathbf{\Sigma}' \cdot \mathbf{V}' = \mathbf{Y}'$$

The diagram illustrates the truncated Singular Value Decomposition (SVD) of matrix $\mathbf{Y}$. Matrix $\mathbf{U}'$ is a 10x4 matrix containing the first 4 columns of $\mathbf{U}$. Matrix $\mathbf{\Sigma}'$ is a 4x4 diagonal matrix containing the first 4 singular values of $\mathbf{\Sigma}$. Matrix $\mathbf{V}'$ is a 10x4 matrix containing the first 4 columns of $\mathbf{V}$. The product of these three matrices equals matrix $\mathbf{Y}'$, which is the truncated version of $\mathbf{Y}$.